

# **SIMATS ENGINEERING ASSESSMENT TOOL 2**

**COURSE CODE: ITA 06**

**COURSE NAME: MACHINE LEARNING**

## **COURSE OUTCOME COVERED**

**CO2:** *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

*Assessment Tool Weightage: 20%*

**QUESTIONS: 15**

**Total Marks:25**

Annexure A

## **SCENARIO-BASED QUESTIONS**

### ***Code of Conduct:***

*I D. Sindhu (192424197) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

*D. Sindhu.*

***Signature of the student:***

## Annexure A

### Short Answer Test

1. With the help of a neat sketch, explain the different functional blocks present in a machine learning system, including data collection, preprocessing, feature extraction, model training, and evaluation. Illustrate the working of the system by providing suitable examples for both supervised and unsupervised learning approaches, clearly highlighting how labelled and unlabelled data are handled within the learning process. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using both historical and real-time data sources. Explain the selection of an appropriate model, including whether a neural network or any other algorithm is suitable for this task, along with justification. Describe the data preprocessing steps and feature selection techniques required to improve model performance. Further, explain the complete process of training, validating, and evaluating the model, and discuss how such a system can assist authorities in early warning, risk assessment, and effective emergency response. (5 Marks)

3. In the context of neural networks, consider a multilayer perceptron used for binary classification. Explain how the backpropagation algorithm updates the weights in the network during training. Illustrate the process using a simple example by describing the forward pass, error calculation, and backward propagation of errors. Discuss how learning rate and activation functions influence the convergence and accuracy of the model. (5 Marks)

4. In the context of Genetic Algorithms, consider a dataset representing different candidate solutions for maximizing a fitness function based on two parameters: Speed and Accuracy. Each solution is evaluated as either suitable (Yes) or not suitable (No). Using the Genetic Algorithm approach, analyze how the population evolves by applying selection, crossover, and mutation operations. Show the step-by-step improvement of candidate solutions across generations and identify the best optimal solution. (5 Marks)

Dataset:

| Example | Speed  | Accuracy | Fitness Value | Suitable Solution |
|---------|--------|----------|---------------|-------------------|
| 1       | High   | Low      | 40            | No                |
| 2       | Medium | Medium   | 60            | Yes               |
| 3       | High   | Medium   | 75            | Yes               |
| 4       | Low    | High     | 65            | Yes               |
| 5       | Medium | High     | 80            | Yes               |

Explain how selection chooses the best individuals, how crossover generates new offspring, and how mutation helps in exploring the hypothesis space. Also, provide the final optimized solution after evolution.

5. In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood

Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. (5 Marks)

Dataset:

| Example | Age | Blood Pressure | Glucose Level | Lifestyle Risk | Actual Output | Predicted Output |
|---------|-----|----------------|---------------|----------------|---------------|------------------|
| 1       | 45  | High           | High          | High           | Yes           | Yes              |
| 2       | 50  | Normal         | High          | Medium         | Yes           | No               |
| 3       | 30  | Normal         | Normal        | Low            | No            | No               |
| 4       | 60  | High           | High          | High           | Yes           | Yes              |
| 5       | 40  | Normal         | Normal        | Medium         | No            | Yes              |

Using the above dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability.

6. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models. (5 Marks)

7. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making. (5 Marks)

8. In the context of **Convolutional Neural Networks (CNNs)**, explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance. (5 Marks)

9. Consider the application of **Genetic Algorithms for feature selection** in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity. (5 Marks)

10. In the context of **classification problems**, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection. (5 Marks)

11. With the help of a neat diagram, explain the architecture of an **Artificial Neural Network (ANN)**. Describe the roles of input, hidden, and output layers, and explain how weights and biases are used in computation. Illustrate how forward propagation works with a simple example, and discuss how overfitting can be controlled using techniques such as dropout and regularization. (5 Marks)

12. Design a machine learning system for **weather forecasting** to predict rainfall using features such as temperature, humidity, wind speed, and atmospheric pressure. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time predictions. (5 Marks)

13. In the context of **Recurrent Neural Networks (RNNs)**, explain the architecture and how they handle sequential data. Describe the concept of hidden states and how information is passed through time steps. Illustrate with an example such as text prediction or time-series forecasting, and discuss challenges like vanishing gradients and possible solutions. (5 Marks)

14. Consider the use of **Genetic Algorithms for optimization problems** such as scheduling tasks in a system. Explain how solutions are encoded as chromosomes and how the fitness function is defined. Describe the processes of selection, crossover, and mutation, and discuss how these operations help in finding near-optimal solutions efficiently. (5 Marks)

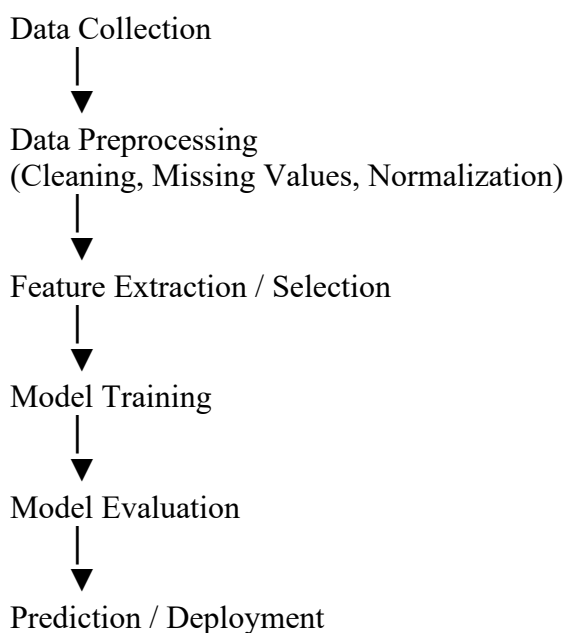
15. In the context of **classification models**, explain how performance evaluation is carried out using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Using a sample confusion matrix, demonstrate how each metric is calculated and discuss their importance in real-world applications such as spam detection and disease diagnosis. (5 Marks)

## RUBRICS FOR CALCULATION

| Criteria                          | Weightage | Level 4 – Exemplary (5 Marks)                                | Level 3 – Proficient (4 Marks)         | Level 2 – Developing (2–3 Marks)             | Level 1 – Beginning (0–1 Mark)  |
|-----------------------------------|-----------|--------------------------------------------------------------|----------------------------------------|----------------------------------------------|---------------------------------|
| <b>Conceptual Understanding</b>   | <b>30</b> | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps     | Basic understanding with some misconceptions | Poor or incorrect understanding |
| <b>Accuracy of Answer</b>         | <b>20</b> | Completely correct answer with no errors                     | Mostly correct with minor errors       | Partially correct with noticeable errors     | Incorrect or irrelevant answer  |
| <b>Application of Knowledge</b>   | <b>20</b> | Correctly applies concepts to solve the question/problem     | Applies concepts with minor mistakes   | Limited or partially correct application     | No or incorrect application     |
| <b>Completeness of Response</b>   | <b>15</b> | Covers all required parts of the question                    | Covers most parts with minor omissions | Covers only some parts                       | Incomplete or missing response  |
| <b>Clarity &amp; Presentation</b> | <b>15</b> | Clear, concise, and well-organized answer                    | Understandable with minor issues       | Somewhat unclear or poorly structured        | Difficult to understand         |

## 1. Functional Blocks of a Machine Learning System

### Neat Diagram



### Explanation

- **Data Collection:** Gather data from databases, sensors, or websites.
- **Preprocessing:** Remove missing values, normalize data, and encode categorical values.
- **Feature Extraction:** Select important features for learning.
- **Model Training:** Train algorithms using the prepared data.
- **Evaluation:** Measure performance using Accuracy, Precision, Recall, and F1-score.

### Example

- **Supervised Learning:** Email Spam Detection (uses **labelled data**: Spam/Not Spam).
- **Unsupervised Learning:** Customer Segmentation using K-Means (uses **unlabelled data**).

## 2. ML System for Disaster Management

### Architecture

Historical + Real-time Data



Preprocessing



Feature Selection



Machine Learning Model  
(Random Forest / Neural Network)



Prediction



Early Warning System

### Model

- **Random Forest:** Good for structured weather data.
- **Neural Network:** Suitable for complex patterns.

### Preprocessing

- Handle missing values.
- Normalize data.
- Remove duplicates.

### **Feature Selection**

- Rainfall
- River level
- Temperature
- Seismic activity

### **Training & Evaluation**

- Split data into Train, Validation, Test.
- Evaluate using Accuracy, Precision, Recall.

### **Applications**

- Flood prediction
- Earthquake risk assessment
- Emergency response
- Public warning systems

## **3. Backpropagation in Neural Networks**

### **Steps**

#### **1. Forward Pass**

- Inputs pass through hidden and output layers.

#### **2. Error Calculation**

$$Error = Target - Output$$

#### **3. Backward Pass**

- Error propagates backward.
- Weights updated using:

$$w_{new} = w + \eta(T - O)x$$

### **Example**

Target =1

Output =0.8

Error =0.2

Weights are updated to reduce future error.

### **Learning Rate**

- Small → Slow learning

- Large → Overshooting

### Activation Functions

- Sigmoid
- ReLU
- Tanh

They improve convergence and model accuracy.

## 4. Genetic Algorithm

### Dataset

| Solution | Fitness |
|----------|---------|
| 1        | 40      |
| 2        | 60      |
| 3        | 75      |
| 4        | 65      |
| 5        | 80      |

### Selection

Best individuals:

- Solution 5 (80)
- Solution 3 (75)

### Crossover

Parents:

10110

11001

Single-point crossover

Offspring

10101

11010

### Mutation

Randomly flip one bit

10101 → 10111

### Final Solution

Solution with highest fitness (**80 or improved offspring**) is selected.

Mutation increases diversity and prevents local optima.



## 5. Neural Network Classification

### Confusion Matrix

| Example | Actual | Predicted |
|---------|--------|-----------|
| 1       | Yes    | Yes       |
| 2       | Yes    | No        |
| 3       | No     | No        |
| 4       | Yes    | Yes       |
| 5       | No     | Yes       |

### TP, TN, FP, FN

TP =2

TN =1

FP =1

FN =1

### Accuracy

$$= \frac{TP + TN}{Total} = \frac{3}{5} = 60\%$$

### Precision

$$= \frac{2}{3} = 66.67\%$$

### Recall

$$= \frac{2}{3} = 66.67\%$$

### F1-score

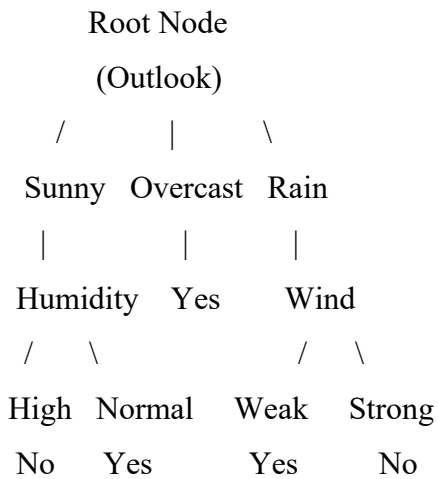
$$66.67\%$$

### Improvements

- Data preprocessing removes noise.
- Feature selection removes irrelevant attributes.

## 6. Decision Tree Learning

### Diagram



### Feature Selection

Uses:

- Entropy
- Information Gain

### Tree Construction

- Select feature with highest Information Gain.
- Split recursively.

### Pruning

Removes unnecessary branches to reduce overfitting.

## 7. Smart Traffic Management System

### Inputs

- Vehicle density
- Time
- Weather
- Road type

### Algorithm

Random Forest or Neural Network.

### Preprocessing

- Remove missing values
- Normalize data
- Encode categorical values

## Training

Train → Validate → Test

## Deployment

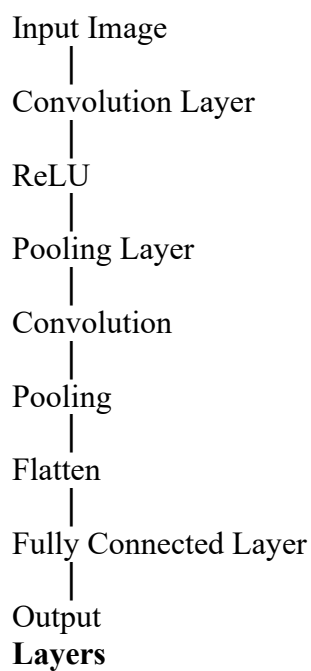
Real-time traffic prediction using sensors and cameras.

Benefits:

- Route optimization
- Traffic control
- Reduced congestion

## 8. CNN Architecture

### Diagram



- **Convolution:** Extracts features.
- **Pooling:** Reduces dimensions.
- **Fully Connected:** Classification.

### Backpropagation

Updates filters using gradient descent.

### Example

Cat vs Dog image classification.

Factors affecting performance:

- Dataset size
- Learning rate
- Number of layers

## 9. Genetic Algorithm for Feature Selection

### Chromosome

[1 0 1 1 0]

1 → Selected feature

0 → Not selected

### Fitness

Accuracy of ML model.

### Operations

- Selection → Best chromosomes
- Crossover → New feature subsets
- Mutation → Random feature changes

### Benefits

- Better accuracy
- Reduced computation
- Lower complexity

## 10. Evaluation Metrics

### Sample Confusion Matrix

TP=90

TN=80

FP=10

FN=20

### Accuracy

$$= \frac{170}{200} = 85\%$$

### Precision

$$= \frac{90}{100} = 90\%$$

### Recall

$$= \frac{90}{110} = 81.82\%$$

### F1-score

$$85.71\%$$

### ROC Curve

Plots:

- True Positive Rate
- False Positive Rate

Applications:

- Disease diagnosis
- Fraud detection

## 11. ANN Architecture

### Diagram

Input Layer

|

Hidden Layer

|

Hidden Layer

|

Output Layer

### Components

- Input Layer → Receives features.
- Hidden Layer → Learns patterns.
- Output Layer → Gives prediction.

### Computation

$$Net = \sum (wx) + b$$

Activation Function:

ReLU/Sigmoid

## Overfitting Control

- Dropout
- L2 Regularization
- Early Stopping

## 12. Weather Forecasting ML System

### Inputs

- Temperature
- Humidity
- Wind speed
- Pressure

### Algorithm

Random Forest or ANN.

### Preprocessing

- Handle missing values
- Normalize data
- Feature engineering

### Training

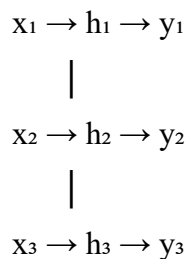
Train → Validation → Testing

### Deployment

Real-time rainfall prediction using weather stations.

## 13. Recurrent Neural Networks (RNN)

### Diagram



### Hidden State

Stores previous information.

### **Example**

- Text prediction
- Weather forecasting
- Stock price prediction

### **Challenges**

- Vanishing gradient
- Exploding gradient

### **Solutions**

- LSTM
- GRU

## **14. Genetic Algorithm for Scheduling**

### **Chromosome**

Task1 Task2 Task3 Task4

### **Fitness**

Measures:

- Minimum completion time
- Maximum resource utilization

### **Operations**

- Selection → Choose best schedules.
- Crossover → Exchange task order.
- Mutation → Swap tasks randomly.

### **Advantages**

- Near-optimal scheduling
- Faster convergence
- Better resource allocation

## **15. Classification Performance Evaluation**

### **Sample Confusion Matrix**

TP=95

TN=85

FP=10

FN=10

### Accuracy

$$= \frac{180}{200} = 90\%$$

### Precision

$$= \frac{95}{105} = 90.48\%$$

### Recall

$$= \frac{95}{105} = 90.48\%$$

### F1-score

$$90.48\%$$

### ROC-AUC

- Measures classifier performance across different thresholds.
- Higher AUC indicates better classification.

### Applications

- **Spam Detection:** High precision reduces false spam classification.
- **Disease Diagnosis:** High recall minimizes missed disease cases.